

Formative R activities: A Small COVID-19 Dataset

1. Objective

This project is a series of exploratory data analysis and statistical activities conducted in R using the *COVID-19 India Cases (January–March 2020)* dataset sourced from Kaggle (Teloo, A. (2021)). The dataset contains early COVID-19 reporting data across Indian states and foreign territories during the initial phase of the pandemic, including confirmed cases among Indian and foreign nationals, recovered cases, and deaths.

The activities focused on developing practical skills in data exploration, descriptive statistics, creating categorical variables, frequency analysis, and data visualisation. Using R, the dataset was examined to identify reporting patterns, case distributions, and statistical characteristics across regions and case categories. The R reporting script is available in Appendix B.

These exercises supported the learning outcomes of developing a systematic understanding of key mathematical and statistical concepts underpinning Data Science and Artificial Intelligence, and applying statistical methods to support analytical reasoning and data-driven decision-making. The work also contributed to developing practical experience in reproducible analytical workflows, interpreting statistical outputs, and visualising data insights.

2. Dataset Overview and Initial Exploration

This section includes the number of observations and variables, variable names and types, unique states/territories, the date range, and the most frequently reported state.

2.1. Dataset Structure

The imported 'covid' dataset has **270 observations and 7 variables**; however, an initial data inspection showed that the **Sno** index was created during the upload and does not add value to this dataset; hence, it has been removed, resulting in 6 variables, of which 4 were in numeric format and 2 in character/text format. As the **Date** variable was not formatted as a date but as a character, it had to be formatted

to the right date format ("%Y-%m-%d"). The final dataset contains 6 variables: 4 numeric, 1 date, and 1 character.

	Sno	Date	State/UnionTerritory	ConfirmedIndianNational	ConfirmedForeignNational	Cured	Deaths
1	1	30-01-2020	Kerala	1	0	0	0
2	2	31-01-2020	Kerala	1	0	0	0
3	3	01-02-2020	Kerala	2	0	0	0
4	4	02-02-2020	Kerala	3	0	0	0
5	5	03-02-2020	Kerala	3	0	0	0
6	6	04-02-2020	Kerala	3	0	0	0
7	7	05-02-2020	Kerala	3	0	0	0
8	8	06-02-2020	Kerala	3	0	0	0
9	9	07-02-2020	Kerala	3	0	0	0
10	10	08-02-2020	Kerala	3	0	0	0
11	11	09-02-2020	Kerala	3	0	0	0
12	12	10-02-2020	Kerala	3	0	0	0
13	13	11-02-2020	Kerala	3	0	0	0

Table 1. Original Covid dataset

```
> str(covid)
'data.frame': 270 obs. of 7 variables:
 $ Sno          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Date         : chr  "30-01-2020" "31-01-2020" "01-02-2020" "02-02-2020" ...
 $ State.UnionTerritory : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 0 ...
 $ Cured        : int  0 0 0 0 0 0 0 0 0 0 ...
 $ Deaths      : int  0 0 0 0 0 0 0 0 0 0 ...
```

	Date	State.UnionTerritory	ConfirmedIndianNational	ConfirmedForeignNational	Cured	Deaths
127	2020-03-13	Telengana	1	0	0	0
128	2020-03-13	Uttar Pradesh	10	1	0	0
129	2020-03-13	Union Territory of Ladakh	3	0	0	0
130	2020-03-13	Tamil Nadu	1	0	0	0
131	2020-03-13	Union Territory of Jammu and Kashmir	1	0	0	0
132	2020-03-13	Punjab	1	0	0	0
133	2020-03-13	Karnataka	6	0	0	1
134	2020-03-13	Maharashtra	14	0	0	0
135	2020-03-13	Andhra Pradesh	1	0	0	0
136	2020-03-14	Delhi	7	0	1	1
137	2020-03-14	Haryana	0	14	0	0
138	2020-03-14	Kerala	19	0	3	0
139	2020-03-14	Rajasthan	1	2	1	0
140	2020-03-14	Telengana	1	0	0	0
141	2020-03-14	Uttar Pradesh	11	1	5	0

Table 2. Cleaned Covid dataset

```

> # 10. Data format final version
> str(covid)
'data.frame': 270 obs. of 6 variables:
 $ Date      : Date, format: "2020-01-30" "2020-01-31" "2020-02-01" "2020-02-02" ...
 $ State.UnionTerritory : chr  "Kerala" "Kerala" "Kerala" "Kerala" ...
 $ ConfirmedIndianNational : int  1 1 2 3 3 3 3 3 3 ...
 $ ConfirmedForeignNational: int  0 0 0 0 0 0 0 0 0 ...
 $ Cured      : int  0 0 0 0 0 0 0 0 0 ...
 $ Deaths    : int  0 0 0 0 0 0 0 0 0 ...
> summary(covid)
      Date      State.UnionTerritory ConfirmedIndianNational ConfirmedForeignNational      Cured
Min.   :2020-01-30 Length   :270 Min.   : 0.000 Min.   : 0.00 Min.   :0.00000
1st Qu.:2020-03-08 N.unique : 27 1st Qu.: 1.000 1st Qu.: 0.00 1st Qu.:0.00000
Median :2020-03-13 N.blank  : 0 Median : 3.000 Median : 0.00 Median :0.00000
Mean   :2020-03-10 Min.nchar: 5 Mean   : 5.552 Mean   : 1.33 Mean   :0.6037
3rd Qu.:2020-03-18 Max.nchar: 36 3rd Qu.: 6.000 3rd Qu.: 0.00 3rd Qu.:0.00000
Max.   :2020-03-21 Max.   :60.000 Max.   :14.00 Max.   :9.00000
Deaths
Min.   :0.00000
1st Qu.:0.00000
Median :0.00000
Mean   :0.09259
3rd Qu.:0.00000
Max.   :1.00000

```

2.2. Data Characteristics

- The COVID data set contains **27 unique states/union territories**
- The data range covered in the data is from **2020-01-30 until 2020-03-21**
- The most frequent cases were represented by **Kerala with 52**

```

> # 8. Date range covered
> range(covid$Date)
[1] "2020-01-30" "2020-03-21"
> min(covid$Date)
[1] "2020-01-30"
> max(covid$Date)
[1] "2020-03-21"
> # -----
>
> # 9. Number of unique states/union territories
> length(unique(covid$State.UnionTerritory))
[1] 27
>
> # 10. Most frequent state
> state_freq <- table(covid$State.UnionTerritory)
> sort(state_freq, decreasing = TRUE)[1]
Kerala
52
..

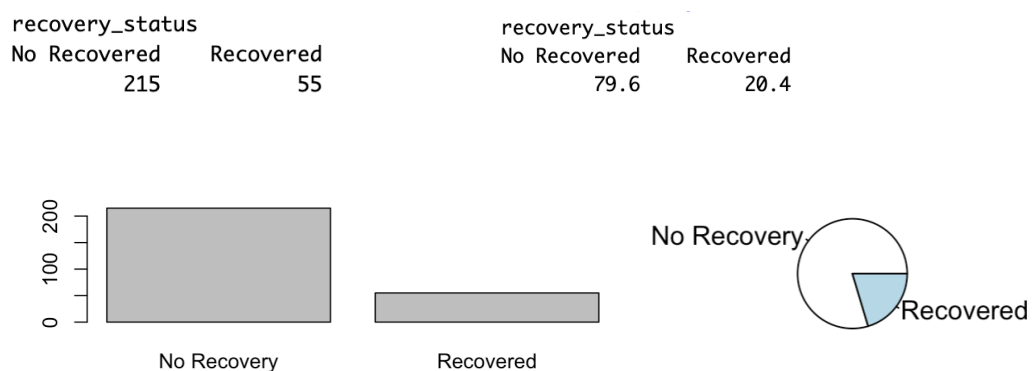
```

3. Frequency Analysis, Binary Variable and Factor Variable

This section provides frequency tables for categorical analysis and reports on recoveries and deaths by creating the `has_recovery` and `has_deaths` variables, and includes bar plots and pie charts.

3.1. Recovered cases

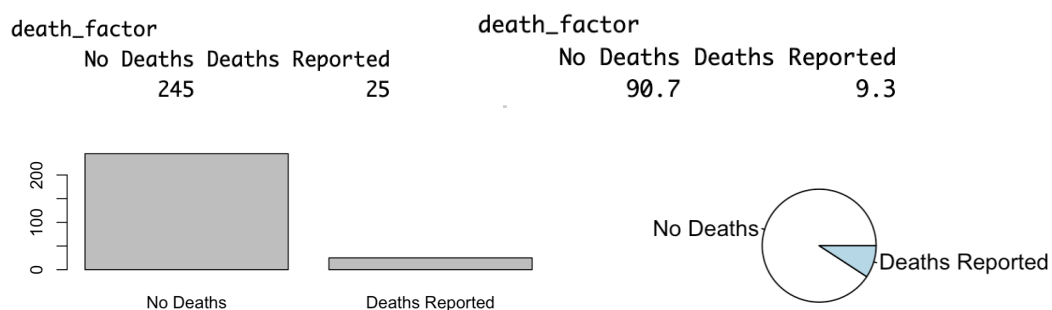
The frequency table for the variable `has_recovery`, which was transformed into a labelled categorical variable for easier interpretation, shows 215 non-recovered vs 55 Recovered cases, with 80% non-recovered.



Figures 1. Bar and plot charts for recovered cases

3.2. Death cases

Frequency tables for the `has_deaths` variable, transformed into a factor, show 245 non-death cases and 25 death cases, with 90% non-death cases.



Figures 2. Bar and plot charts for death cases

4. Descriptive Statistics and Measures of Dispersion

This section includes **information on Central tendency** (mean, median, mode) and **Dispersion** (variance, standard deviation, and range) **across Indian states** ([ConfirmedIndianNational](#)), along with corresponding visualisations: histograms with density curves for each variable, boxplots to identify outliers, and violin plots to show the shape of the distribution.

4.1. Central tendency and Dispersion for [ConfirmedIndianNational](#)

- **Central tendency:** “*What is typical?*”
 - **Mean:** average number of confirmed cases: **5.6**
 - **Median:** middle value of cases: **3**
 - **Mode:** most common case count: **1 occurs most often**
- **Dispersion:** “*How spread out is the data?*”
 - **Range:** **60** (min to max cases: **0-60**)
 - **Variance:** measures how close the numbers are to the mean (*defined as the average squared difference of each number from the mean*): **69.5**
 - **Standard deviation** (sd): measures how much individual numbers differ on average, from the mean (*defined as squared root of variance*): **8.3**

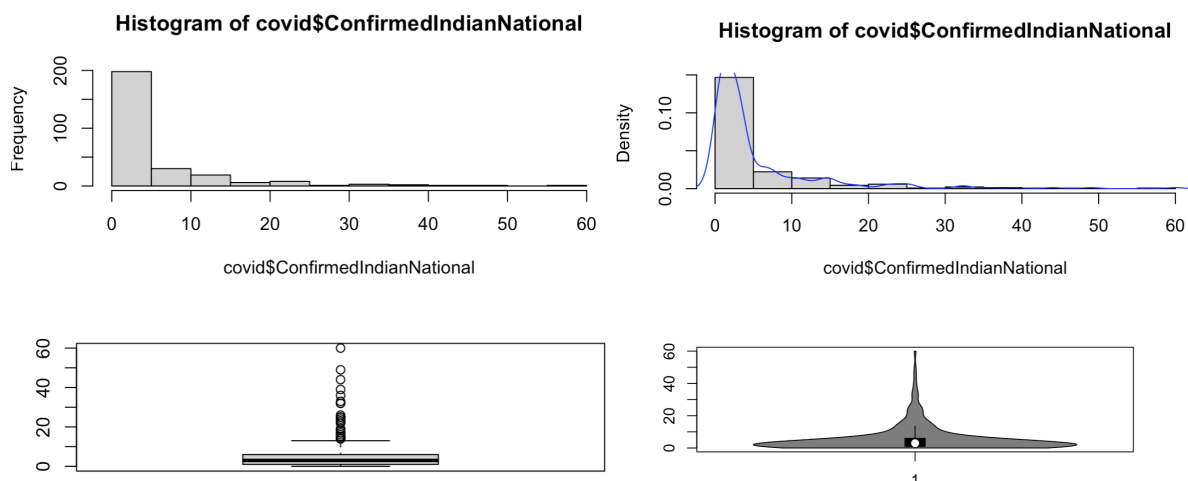


Figure 3. Histograms with density curves, boxplots and violin plots for `ConfirmedIndianNational` cases

The measures of central tendency indicate that the typical number of confirmed Indian national Covid cases per observation was relatively low during the analysed period. While the average number of confirmed cases was 5.6, the median value of 3 and the mode of 1 suggest that most observations contained only a small number of cases. The mean is notably higher than both the median and the mode, indicating a positively skewed distribution because a small number of observations with high case counts raise the overall average.

The measures of dispersion show substantial variability in the dataset. The large variance of 69.5 and the standard deviation of 8.3 were higher than the mean of 5.6, indicating that case numbers varied considerably across observations. The wide range of 0 to 60 further confirms the presence of both very low and relatively high case counts within the dataset. Overall, although most observations reported low case counts, a smaller number reported significantly higher counts, contributing to a positively (right)- skewed distribution with notable outliers in the visualisations.

4.2. Visualisation for other variables

The visualisations across the four variables (**ConfirmedIndianNational**, **ConfirmedForeignNational**, **Cured**, and **Deaths**) showed that all distributions were positively skewed, with most observations concentrated at low values and a smaller number of observations containing much higher counts. The boxplots revealed multiple outliers, particularly for confirmed cases and recoveries, indicating variability across observations and states. Histograms and violin plots further highlighted that deaths were heavily concentrated near zero, while confirmed cases displayed the greatest spread and variability. Overall, the visualisations demonstrated that the early Covid data were highly unevenly distributed, with a few observations contributing disproportionately higher case counts.

4.2.1. ConfirmedForeignNational

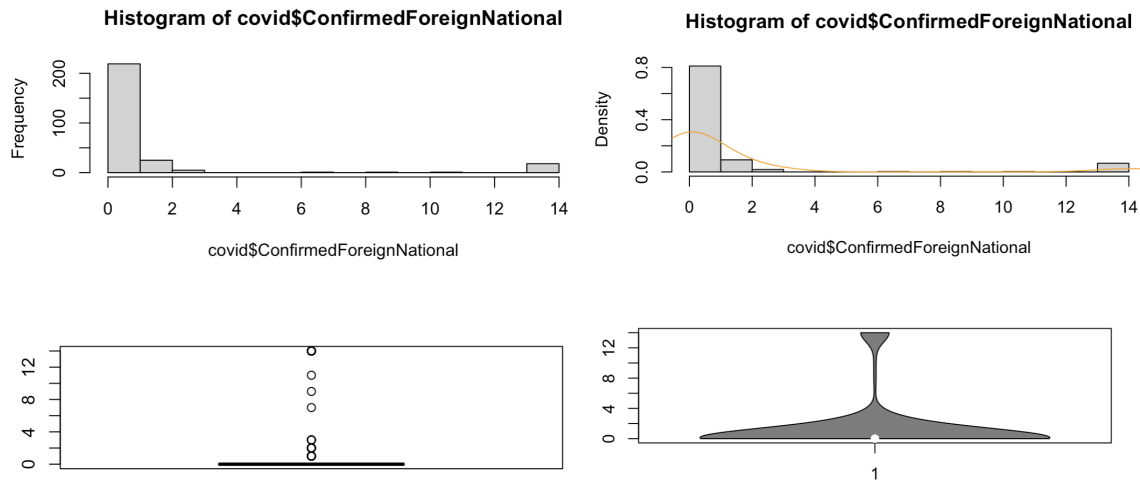


Figure 4. Histograms with density curves, boxplots and violin plots for ConfirmedForeignNational cases

4.2.2. Cured

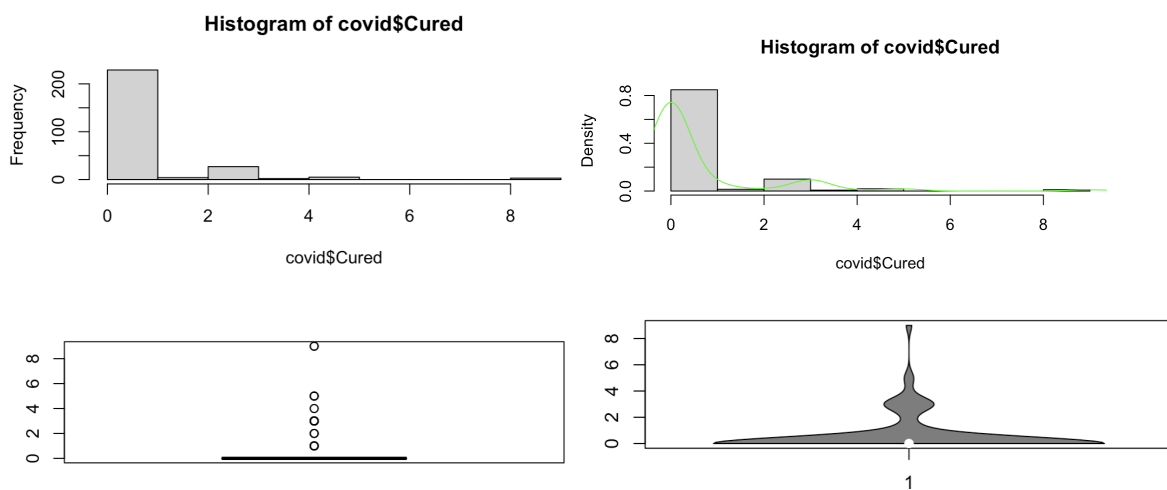


Figure 5. Histograms with density curves, boxplots and violin plots for Cured cases

4.2.3. Deaths

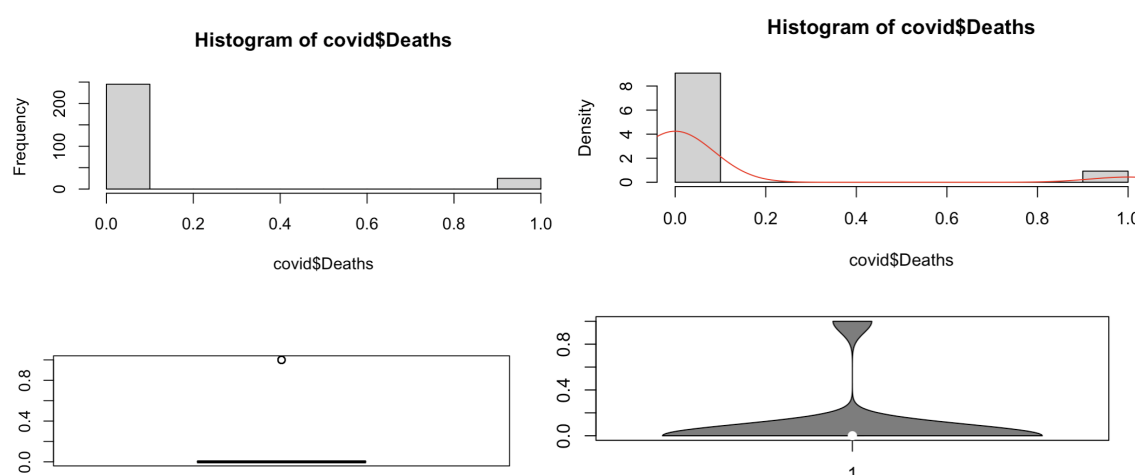


Figure 6. Histograms with density curves, boxplots and violin plots for Deaths cases

5. Categorical Factor Variables and Contingency Table

This section reports on the total_cases variable, case_level categories, factor levels and frequency table for the State/Union Territory variable and identifies the top 10 states with the highest number of daily reports.

5.1. Total cases and case severity variables

Total cases variable combined Indian and Foreign nationals data, and **case severity categorical variables** (No Cases, Low Cases, Medium Cases, High Cases) were defined based on the predefined criteria, resulting in 177 “low cases”, 61 “medium cases”, 32 “high cases” and 0 “no cases” reported.

Predefined criteria	Result
No Cases = 0	0
Low Cases = 1-5 cases	177
Medium Cases = 6-15 cases	61
High Cases - 16+ cases	32

Table 1. Covid severity results

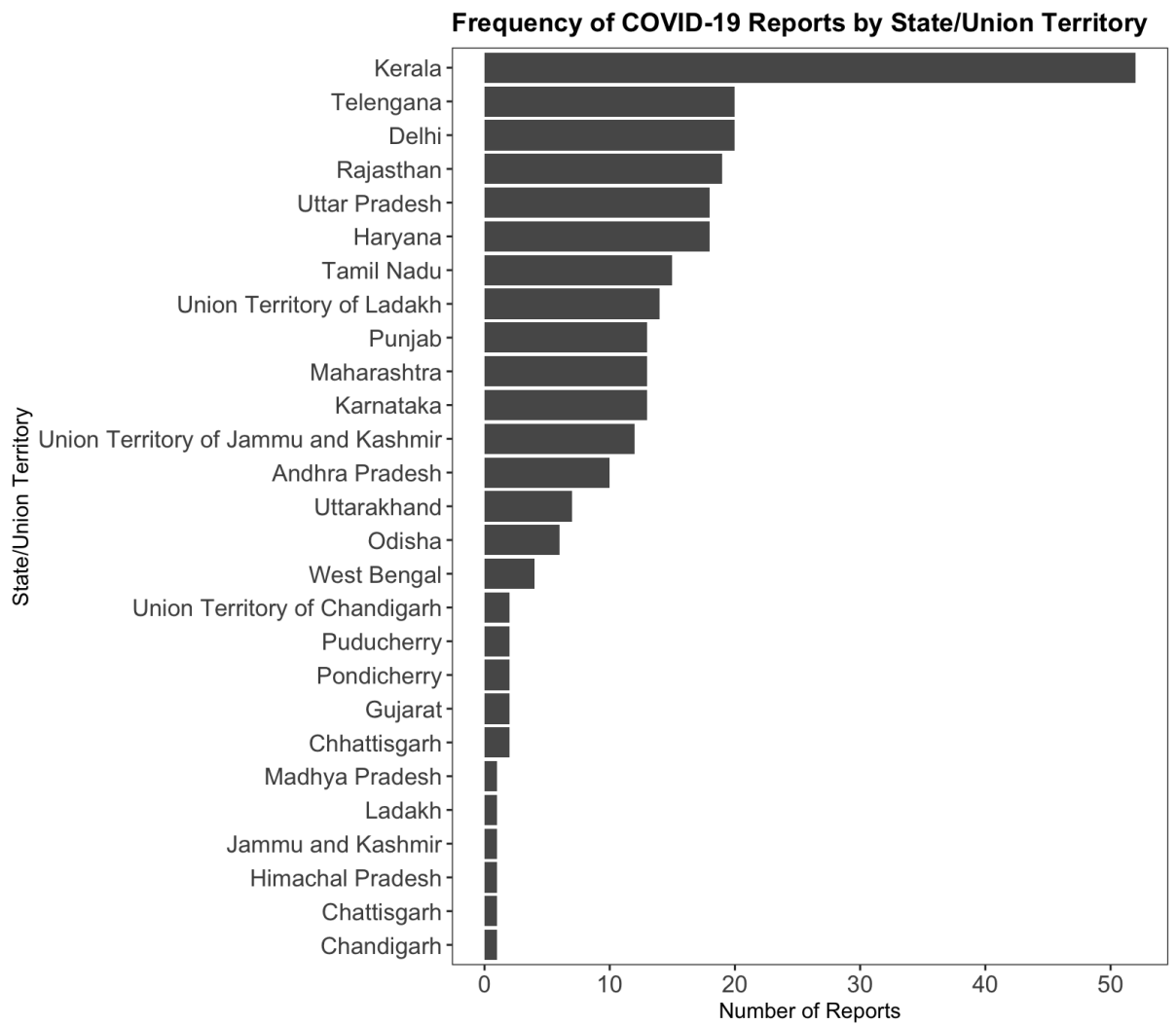
5.2. Top 10 states with the highest number of daily reports

The rank of the reported cases was led by Kerala, with 52 cases (11 high, 3 medium, and 38 low), followed by Karnataka and Maharashtra, with 13 each, rounding out the list.

Kerala	Delhi	Telangana	Rajasthan
52	20	20	19
Haryana	Uttar Pradesh	Tamil Nadu	Union Territory of Ladakh
18	18	15	14
Karnataka	Maharashtra		
13	13		

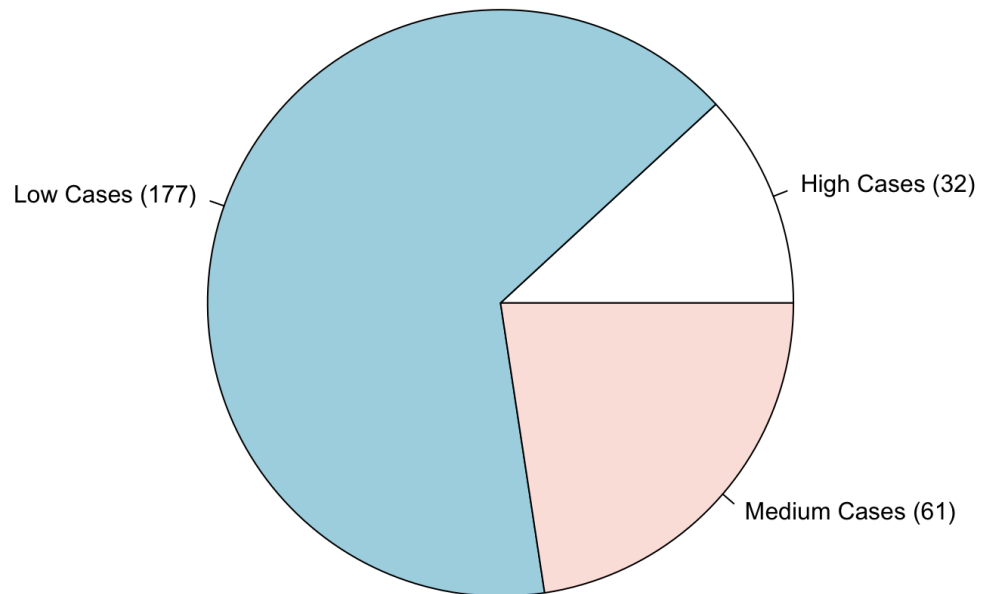
	case_level				Sum
	No Cases	Low Cases	Medium Cases	High Cases	
Kerala	0	38	3	11	52
Delhi	0	10	8	2	20
Telangana	0	16	2	2	20
Rajasthan	0	9	8	2	19
Haryana	0	7	7	4	18
Uttar Pradesh	0	0	14	4	18
Tamil Nadu	0	15	0	0	15
Union Territory of Ladakh	0	10	4	0	14
Karnataka	0	4	9	0	13
Maharashtra	0	3	3	7	13
Sum	0	112	58	32	202

6. Frequency of reported cases by State/Union Territory



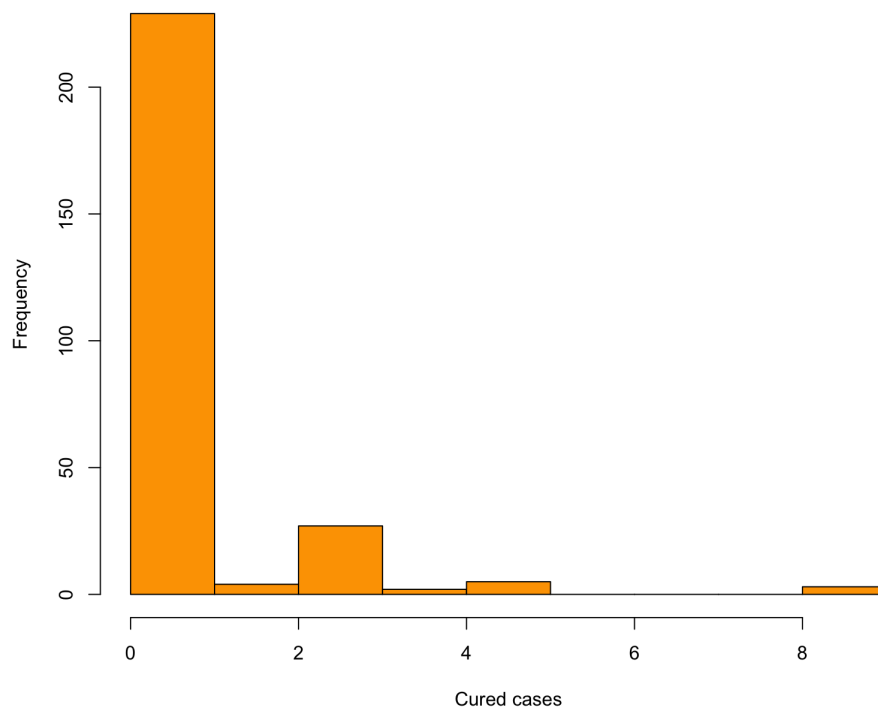
7. Pie chart: Distribution of case severity levels based on total confirmed cases

Pie Chart of Distribution of COVID-19 Case Severity Levels

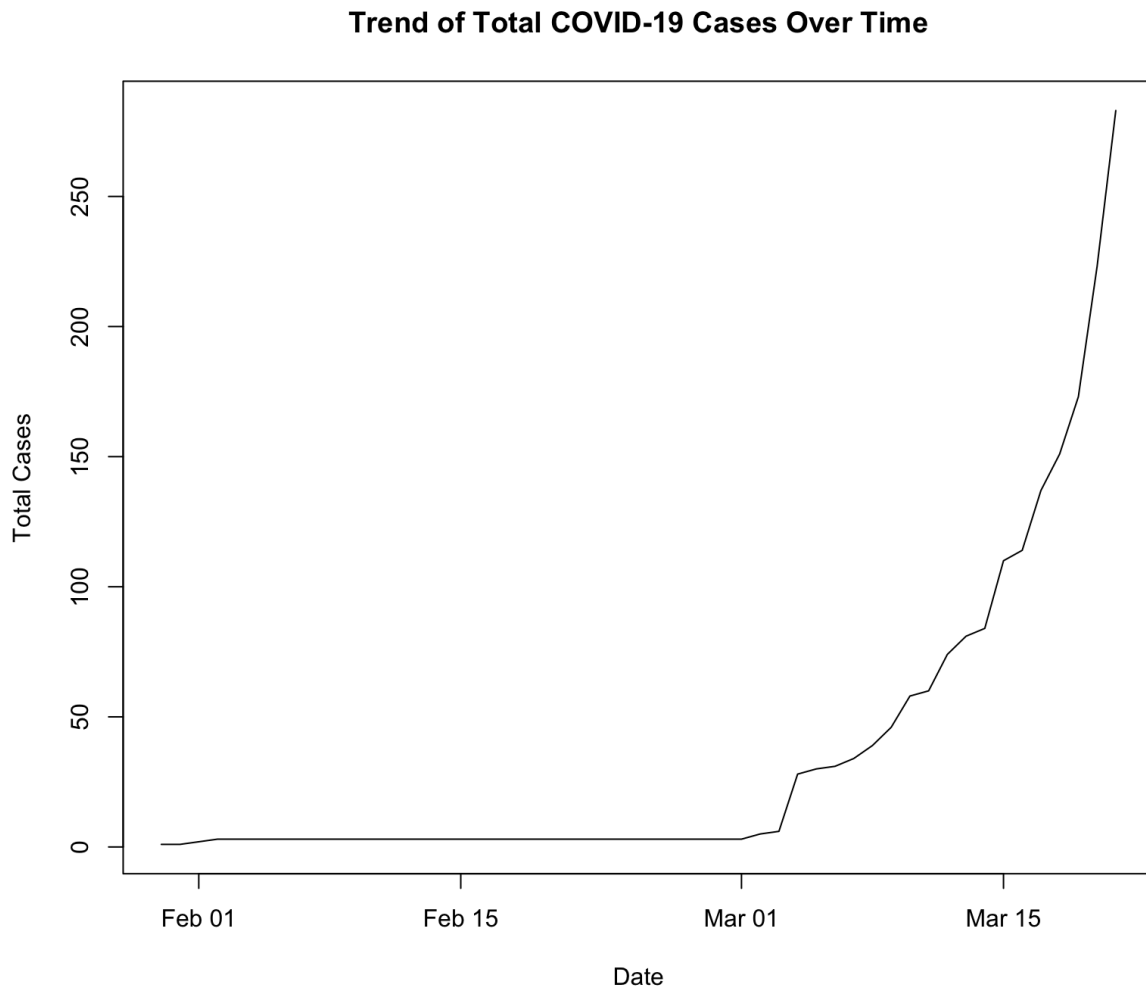


8. Histogram with the distribution of recovery numbers

Distribution of Recovery Numbers of COVID-19



9. Line chart showing the trend of total cases over time



10. Key Learnings and Reflection

Through these activities, I have developed a more structured and reproducible workflow in R by moving from ad hoc console experimentation to more organised, script-based analysis. I have learned the importance of maintaining clean, reusable scripts, separating raw and cleaned datasets, and organising project folders to support reproducibility and facilitate debugging.

The exercises also highlighted the importance of data cleaning before analysis, particularly when handling issues such as incorrect date formats, unnecessary index variables, and inconsistent variable types.

In addition, I gained practical experience working with factors, categorical variables, and frequency and contingency tables to analyse reporting patterns across states and case categories, and to sort and summarise grouped data.

By calculating descriptive statistics and creating visualisations such as histograms, boxplots, and violin plots, I improved my understanding of measures of central tendency, dispersion, skewness, and outliers. Moreover, data trends are presented using a line chart and the FUN function to aggregate data by date, and distributions are shown with the ggplot library, pie charts, and histograms.

Overall, the activities helped me move from having no R knowledge and basic analysis to a more systematic, analytical approach to data exploration and interpretation.

Reference

Teloo, A. (2021) *A Small COVID-19 Dataset*. Kaggle. Available at: <https://www.kaggle.com/datasets/aditeloo/a-small-covid19-dataset> (Accessed: 15 May 2026)

Appendix A - Data and Values created

Data	
 covid	270 obs. of 6 variables 
Values	
case_level	Factor w/ 4 levels "No Cases","Low Cases",...: 2 2 2 ...
death_factor	Factor w/ 2 levels "No Deaths","Deaths Reported": 1 ...
frequency1	'table' int [1:2(1d)] 215 55
has_deaths	logi [1:270] FALSE FALSE FALSE FALSE FALSE FALSE ...
has_recovery	logi [1:270] FALSE FALSE FALSE FALSE FALSE FALSE ...
recovery_status	chr [1:270] "No Recovered" "No Recovered" "No Recove...
state_case_table	'table' int [1:27, 1:4] 0 0 0 0 0 0 0 0 0 0 ...
state_freq	'table' int [1:27(1d)] 10 1 1 2 20 2 18 1 1 13 ...
top_10_frequency_t...	'table' int [1:10, 1:4] 0 0 0 0 0 0 0 0 0 0 ...
total_cases	int [1:270] 1 1 2 3 3 3 3 3 3 3 ...

Appendix B - R-Script

```
# =====  
# COVID ANALYSIS: Activity 1 (Unit 1 /Seminar 1)  
# =====
```

```
# 1. Clear workspace  
rm(list = ls())
```

```
# -----
```

```
# 2. Load packages  
# -----  
library(psych)  
library(vioplot)  
library(skimr)  
# -----
```

```
# 3. Import CSV
```

```

covid <- read.csv("data/covid_india.csv")

# -----

# 4. Inspect data structure
View(covid)
str(covid)
summary(covid)
names(covid)
# -----

# 5. Remove 'Sno' ID
head(covid$Sno)
covid$Sno <- NULL
ncol(covid)
# -----

# 6. Data format and variable type check
sapply(covid, class)
sum(sapply(covid, is.numeric))
sum(sapply(covid, is.character))
# -----

# 7. Convert date field from character to date format
covid$Date <- as.Date(covid$Date, format = "%d-%m-%Y")
class(covid$Date)
head(covid$Date)

# 8. Date range covered
range(covid$Date)
min(covid$Date)
max(covid$Date)
# -----

# 9. Number of unique states/union territories
length(unique(covid$State.UnionTerritory))

# 10. Most frequent state
state_freq <- table(covid$State.UnionTerritory)
sort(state_freq, decreasing = TRUE)[1]
# -----

# 10. Data format final version
str(covid)

```

```

summary(covid)
skim(covid)
# -----

# 11. Save clean dataset
write.csv(covid, "data/covid_india_clean.csv", row.names = FALSE)

# =====
# COVID ANALYSIS: Activity 2 (Unit 2)
# =====

# -----
# 1. Binary variable (has_recovery) & Frequency table
has_recovery <- covid$Cured > 0
table(has_recovery)
# -----

# 2. Recovery status
recovery_status <- ifelse(covid$Cured > 0, "Recovered", "No Recovered")
table(recovery_status)
# -----

# 3. Percentages of recovery cases
frequency1 <- table(recovery_status)
frequency1/sum(frequency1)*100
round(frequency1/sum(frequency1)*100,1)
# -----

# 4. Visualisation
barplot(table(recovery_status))
pie(table(recovery_status))

# =====
# COVID ANALYSIS: Activity 2 (Unit 2/Seminar 2)
# =====

# 1. Central tendency
# -----
mean(covid$ConfirmedIndianNational)
median(covid$ConfirmedIndianNational)
names(sort(table(covid$ConfirmedIndianNational), decreasing = TRUE))[1]
# -----

# 9. Dispersion
# -----

```

```

sd(covid$ConfirmedIndianNational)
var(covid$ConfirmedIndianNational)
range(covid$ConfirmedIndianNational)
# -----
# 10. Visualizations
# -----
hist(covid$ConfirmedIndianNational)
hist(covid$ConfirmedIndianNational, probability = TRUE)
lines(density(covid$ConfirmedIndianNational), col = "blue")
boxplot(covid$ConfirmedIndianNational)
vioplot(covid$ConfirmedIndianNational)

# -----
hist(covid$ConfirmedForeignNational)
hist(covid$ConfirmedForeignNational, probability = TRUE)
lines(density(covid$ConfirmedForeignNational), col = "orange")
boxplot(covid$ConfirmedForeignNational)
vioplot(covid$ConfirmedForeignNational)

# -----
hist(covid$Deaths)
hist(covid$Deaths, probability = TRUE)
lines(density(covid$Deaths), col = "red")
boxplot(covid$Deaths)
vioplot(covid$Deaths)

# -----
hist(covid$Cured)
hist(covid$Cured, probability = TRUE)
lines(density(covid$Cured), col = "green")
boxplot(covid$Cured)
vioplot(covid$Cured)

# =====
# COVID ANALYSIS: Activity 3 (Unit 3)
# =====

# 1. Binary variable (has_deaths) & Frequency table
has_deaths <- covid$Deaths > 0
table(has_deaths)
# -----

# 2. Factor variable with labels ("No Deaths", "Deaths Reported").

```

```
death_factor <- factor(has_deaths, levels = c(FALSE, TRUE), labels = c("No Deaths",  
"Deaths Reported"))  
table(death_factor)  
# -----
```

```
# 3. Percentage of deaths cases  
frequency2 <- table(death_factor)  
frequency2/sum(frequency2)*100  
round(frequency2/sum(frequency2)*100,1)
```

```
# 4. Visualisation  
barplot(table(death_factor))  
pie(table(death_factor))  
# -----
```

```
# 5. Categorical Variable for confirmed cases (Indian + Foreign nationals)  
total_cases <- covid$ConfirmedIndianNational + covid$ConfirmedForeignNational  
case_level <- ifelse(total_cases == 0, "No Cases",  
                     ifelse(total_cases <= 5, "Low Cases",  
                             ifelse(total_cases <= 15, "Medium Cases", "High Cases"))  
# -----
```

```
# 6. Frequency table for the most frequent reporting  
table(case_level)
```

```
case_level <- factor(case_level, levels = c("No Cases", "Low Cases", "Medium  
Cases", "High Cases"))  
table(case_level)
```

```
# -----
```

```
# 7. Frequency table for State/Union Territory for the most frequent reporting  
table(covid$State.UnionTerritory)  
table(covid$State.UnionTerritory, case_level)  
addmargins(table(covid$State.UnionTerritory, case_level))
```

```
# -----
```

```
# 8. Top 10 states with the highest number of daily reports in the dataset  
head(sort(table(covid$State.UnionTerritory), decreasing = TRUE), 10)
```

```
state_case_table <- table(covid$State.UnionTerritory, case_level)  
top_10_frequency_table <- state_case_table[order(rowSums(state_case_table),  
decreasing = TRUE)[1:10],, drop = FALSE]  
addmargins(top_10_frequency_table)
```

```

# -----
# 9. Barchart with frequency of COVID-19 reports by state/union territory.
View(covid_india_clean)

library(tidyverse)

covid_india_clean %>%
  count(State.UnionTerritory) %>%
  ggplot(aes(x = reorder(State.UnionTerritory, n), y = n))+
  geom_col()+
  coord_flip()+

  theme_bw()+
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        legend.text = element_text(size = 12),
        plot.title = element_text(face = "bold"))+

  labs(title = "Plot: Frequency of COVID-19 Reports by State/Union Territory",
        x = "State/Union Territory",
        y = "Number of Reports")

# -----
# 10. Pie chart showing the distribution of case severity levels based on total confirmed
cases
# Using variables ConfirmedIndianNational and ConfirmedForeignNational.

total_cases <- covid_india_clean$ConfirmedIndianNational +
covid_india_clean$ConfirmedForeignNational
case_level <- ifelse(total_cases == 0, "No Cases",
                    ifelse(total_cases <= 5, "Low Cases",
                          ifelse(total_cases <= 15, "Medium Cases", "High Cases")))
table(case_level)

slices <- c(32, 177, 61)
severity_levels <- c("High Cases (32)", "Low Cases (177)", "Medium Cases (61)")
pie(slices, labels = severity_levels, main="Pie Chart of Distribution of COVID-19 Case
Severity Levels")

# -----
# 11. Histogram showing the distribution of recovery numbers

hist(covid_india_clean$Cured,
     main="Distribution of Recovery Numbers of COVID-19",
     xlab="Cured cases",
     col="orange")

```

```
# -----
```

```
# 12. Line chart showing the trend of total cases over time
```

```
covid_india_clean$total_cases <-  
  covid_india_clean$ConfirmedIndianNational +  
  covid_india_clean$ConfirmedForeignNational
```

```
View(covid_india_clean)
```

```
#Aggregate the total cases per Date
```

```
covid_daily_total <- aggregate(total_cases ~ Date, data = covid_india_clean,  
  FUN = sum)
```

```
plot(covid_daily_total$Date, covid_daily_total$total_cases,  
  type = "l",  
  xlab = "Date",  
  ylab = "Total Cases",  
  main = "Trend of Total COVID-19 Cases Over Time")
```

```
# -----
```